

JEE MAIN REVISION MASTER: SIMPLE HARMONIC MOTION SYLLABUS REDUCED

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 171 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Simple Harmonic Motion - Core Application	171 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
-----------------	--

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Simple Harmonic Motion - Core Application (Very High Yield — 171 PYQs)

Core Principle: Key problem pattern in Simple Harmonic Motion.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "The frequency of oscillation of a mass m suspended by a spring is ν_1 . If the length of the spring is cut to half, the same mass osci..." [Ans: C]
- [2026 Evening Shift] Q#1 (MCQ): "A uniform disc of radius R and mass M is free to oscillate about the axis A as shown in the figure. For small oscillations the time pe..." [Ans: A]
- [2026 Morning Shift] Q#1 (MCQ): "A spring of force constant 15N/m is cut into two pieces. If the ratio of their length is $1:3$, then the force constant of smaller piece ..." [Ans: C]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 8th April Evening Shift (Q#1)

The frequency of oscillation of a mass m suspended by a spring is ν_1 . If the length of the spring is cut to half, the same mass oscillates with frequency ν_2 . The value of ν_2/ν_1 is ____.

- (A) 1
- (B) 2
- (C) $\sqrt{2}$ [Correct]
- (D) $\sqrt{3}$

Correct Answer: C

Solution Analysis: When a mass m is attached to a spring with spring constant k , the restoring force is given by Hooke's law $F = -kx$. According to Newton's Second Law, $m \frac{d^2x}{dt^2} = -kx \Rightarrow \frac{d^2x}{dt^2} + (k/m)x = 0$. Comparing this to the standard SHM equation $\frac{d^2x}{dt^2} + \omega^2 x = 0$, we find the angular frequency as, $\omega = \sqrt{(k/m)}$. Since frequency $\nu = (\omega / 2\pi)$, the frequency of oscillation is: $\nu = (1 / 2\pi) \sqrt{(k/m)} \Rightarrow \nu \propto \sqrt{k}$. A spring constant k is inversely proportional to the natural length L of the spring.